

The Intensive–Extensive Hinge

What changes when an otherwise identical physical system is enlarged, divided or combined?

Working rule: Imagine duplicating the system without changing its local state. An *extensive* quantity adds with system size; an *intensive* quantity remains unchanged after equilibrium is restored.

Name: _____

Class: _____

Hinge worksheet series

1. Matter sample: mass, volume and density

Two separate containers each hold 0.75 kg of the same liquid at 20°C. Each sample occupies $6.0 \times 10^{-4} \text{ m}^3$. The samples are combined without loss.

A. Predict before calculating

For the combined sample, predict what happens to mass, volume, density and temperature. Mark each as *doubles*, *halves*, or *unchanged*, and justify your choices.

B. Test the ratio

Calculate the density of one sample and of the combined sample.

$$\rho = \frac{m}{V}.$$

Explain why both numerator and denominator can double while the ratio remains unchanged.

C. Reverse the operation

A uniform sample is divided into three equal parts. State how m , V , ρ and T change for each part. Which conclusion depends on the division being uniform and the system already being in equilibrium?

D. Interrogate the classification

Why is “large numerical value” unrelated to “extensive”? Give one extensive quantity that could have a small value and one intensive quantity that could have a large value.

2. Ideal gas: amount, state and concentration

Two identical rigid vessels each contain 0.40 mol of an ideal gas at 300 K and $1.00 \times 10^5 \text{ Pa}$. A partition is removed so that the gases occupy the combined volume. No energy is exchanged with the surroundings.

A. Represent the before and after states

Sketch the two initial vessels and the final combined vessel. Label n , V , T and p . Identify which quantities add when the system boundary is enlarged.

B. Use the equation of state

Use

$$pV = nRT$$

to show that doubling both n and V at fixed T leaves p unchanged. State what happens to the number density n/V .

C. Separate state from amount

Classify n , V , internal energy U , T , p and n/V as intensive or extensive for this idealised duplication. Explain why U doubles while T does not.

D. Challenge the assumptions

Would the final temperature necessarily remain 300 K if the two initial gases had different temperatures? What additional process would have to occur?

3. Capacitors: stored charge, energy and voltage

Two identical capacitors, each of capacitance $C = 220 \mu\text{F}$, are independently charged to 12 V. They are then connected in parallel with like terminals together. Assume ideal connections and negligible radiation or resistance.

A. Calculate one capacitor

For one capacitor calculate stored charge and energy:

$$Q = CV, \quad E = \frac{1}{2}CV^2.$$

State which quantity describes the electrical state across its terminals.

B. Combine identical systems

Find the equivalent capacitance, total charge, common voltage and total stored energy after connection. Identify which quantities double and which remain unchanged.

C. Explain the physical architecture

Why can charge and stored energy add while voltage remains fixed? Relate this to the addition of conducting area or parallel storage capacity rather than to a change in local electric state.

D. State the limitation

What would happen if the capacitors began at different voltages? Why would the simple “extensive quantities add, intensive quantities stay fixed” rule require an energy-loss qualification?

Synthesis: classify a quantity by a physical operation, not by its name or magnitude

	System	Extensive quantities	Intensive quantities	Operation used to test
1. Complete the comparison.	Liquid sample			
	Ideal gas			
	Capacitors			
		2. Develop the central claim.		
		Respond to:		
		“Extensive and intensive are scaling behaviours under a specified change of system boundary.”		
		Use the terms <i>system boundary</i> , <i>equilibrium</i> , <i>additive</i> , <i>ratio</i> , <i>state variable</i> , <i>amount of substance</i> , and <i>local condition</i> .		

Teacher guide and indicative answers

The Intensive–Extensive Hinge

1. Matter sample

For one sample,

$$\rho = \frac{0.75}{6.0 \times 10^{-4}} = 1.25 \times 10^3 \text{ kg m}^{-3}.$$

After combining,

$$m = 1.50 \text{ kg}, \quad V = 1.20 \times 10^{-3} \text{ m}^3,$$

so

$$\rho = \frac{1.50}{1.20 \times 10^{-3}} = 1.25 \times 10^3 \text{ kg m}^{-3}.$$

Mass and volume are extensive: each adds when identical samples are combined. Density is intensive because it is a local ratio of two extensive quantities that scale together. Temperature remains 20°C because the samples begin in the same thermal state. On division into three equal uniform parts, each part has $m/3$ and $V/3$, while ρ and T remain unchanged. Uniformity and equilibrium matter: a stratified liquid or a sample with a temperature gradient would not give every part the same intensive variables. The terminology concerns scaling, not numerical size. A dust grain may have a tiny extensive mass; a plasma may have an enormous intensive temperature.

2. Ideal gas

Initially, each vessel satisfies

$$pV = nRT.$$

After removing the partition,

$$n_f = 2n, \quad V_f = 2V, \quad T_f = T.$$

Therefore

$$p_f = \frac{(2n)RT}{2V} = \frac{nRT}{V} = p.$$

The number density is likewise unchanged:

$$\frac{n_f}{V_f} = \frac{2n}{2V} = \frac{n}{V}.$$

For this duplication, n , V and U are extensive. Temperature, pressure and number density are intensive. For a monatomic ideal gas,

$$U = \frac{3}{2}nRT,$$

so doubling n at unchanged T doubles U . Temperature represents mean energy per particle, while internal energy represents the total microscopic energy. If initial temperatures differ, energy is redistributed until thermal equilibrium is reached. The final temperature depends on particle numbers, heat capacities and energy conservation; it cannot be obtained by merely declaring temperature intensive.

3. Capacitors

For one capacitor,

$$Q = CV = (220 \times 10^{-6})(12) = 2.64 \times 10^{-3} \text{ C},$$
$$E = \frac{1}{2}CV^2 = \frac{1}{2}(220 \times 10^{-6})(12)^2 = 1.584 \times 10^{-2} \text{ J}.$$

For two identical capacitors in parallel,

$$C_{\text{eq}} = 2C = 440 \mu\text{F},$$
$$Q_{\text{total}} = 2Q = 5.28 \times 10^{-3} \text{ C}, \quad V_f = 12 \text{ V},$$
$$E_{\text{total}} = 2E = 3.168 \times 10^{-2} \text{ J}.$$

Capacitance, charge and stored energy add in this parallel duplication. Voltage, the potential difference across each branch, remains unchanged. If the initial voltages differ, charge flows until a common voltage is reached. Some initially stored electrostatic energy is transferred to thermal energy, radiation or circuit inductance. Thus the simple scaling test applies cleanly only when identical systems begin in the same intensive state and are combined without producing irreversible equilibration.

Teacher synthesis: the classification is conditional on the operation and equilibrium state				
	System	Extensive	Intensive	Qualification
Indicative comparison.	Liquid	m, V	ρ, T	uniform samples at same T
	Gas	n, V, U	$p, T, n/V$	same equilibrium state
	Capacitors	C, Q, E	V	identical voltage before parallel connection
Core interpretation. Intensive and extensive are behaviours under scaling of a system, not permanent labels detached from context. The clean test duplicates an equilibrium system and joins the copies without changing their local state. Additive totals scale; local state variables do not. Ratios of extensive quantities are often intensive, as with $\rho = m/V$ and number density n/V, but this is a useful pattern rather than a substitute for physical reasoning. Geometry and coupling can complicate classifications: capacitance adds in parallel, whereas resistance does not obey one universal size-scaling rule unless the geometry of replication is specified. Expected conclusion. Students should state the operation, boundary and equilibrium assumptions whenever they classify a quantity.				